

Topography Covariate Scaling and COG Conversion Report

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Executive Summary

This report documents the scaling and Cloud Optimized GeoTIFF (COG) conversion of 111 topographic covariate rasters for the AKVEG project.

- **Goal:** Optimize storage and cloud performance while maintaining geomorphometric precision.
- **Result:** 76.8% storage reduction (9.8 TB to 2.3 TB).
- **Integrity:** 100% mask integrity verified via point-by-point terrestrial scaling analysis.

Data Locations

- **Google Earth Engine (GEE):**
 - **Scaled Collection:** projects/akveg-map/assets/covariates/aksdb/aksdb_topo_v20250422_scaled
 - **Raw Collection:** projects/akveg-map/assets/covariates/aksdb/aksdb_topo_v20250422_raw
- **Google Cloud Storage (GCS):**
 - **Production COGs:** gs://akveg-data/aksdb_dem_covars_v20250422_scaled_cogs/
 - **Raw Source:** gs://akveg-data/aksdb_dem_covars_v20250422/
- **Local Configuration:** 03_data_topography/cog_pipeline/scaling_config.json

Naming References

Vetted against the primary project references: 1. **Git Reference:** Alaska Soil Data Bank Project 3 (File: covariates - View 1 (1).tsv) 2. **Email Reference:** 2024-01-04 email '10 m DEM for Alaska' from Colby Brungard (File: Covariate Table2.xlsx).

Processing Implementation

Correction of Background Artifacts

Source data for fluvial metrics contained non-standard negative artifacts in background/ocean areas. The pipeline standardized these to clean NoData values: * **Int16:** -32768 * **Float32:** -99999.0 * **Byte:** 0

Cloud Optimization

Feature	Setting
Tiling	512x512 blocks
Compression	DEFLATE / PREDICTOR=2
Overviews	9 levels (AVERAGE/MODE)

Scaling Strategy Review

Factors were optimized to balance precision vs. dynamic range.

Optimized Scaling Results

- **Curvatures/Openness (100,000 to 1,000,000):** Captured sharp physical gradients at ridges while reducing clamping to < 0.5% for most bands.

- **Local Deviance (1,000):** Expanded range to +/- 32m, capturing previously flattened slopes in rugged terrain.
- **Fluvial Metrics (10.0):** Resolved “staircase” stepping artifacts in `dfa` and `spi`.
- **Categorical (Byte):** Preserved directly as 8-bit integers (0-255) with no scaling to maintain class integrity.

Critical Assessment & Trade-offs

- **Quantization:** Excellent. Most geomorphometric bands show a **Step/IQR < 0.1%**, ensuring that the rounding error is an order of magnitude smaller than the natural variation in the landscape.
- **Hydrological Extremes:** For `spi` (Stream Power Index) and `dfa` (Directional Flow Accumulation), a clamping rate of ~3-6% was accepted. To avoid clamping these extreme values into Int16 space, we would have needed a scale factor near 0.01, which would have destroyed all meaningful precision in 94% of the terrestrial landscape. The current strategy prioritizes precision where 95% of modeling occurs.
- **Fluvial Healing:** The source data for `dfa` and `spi` contained large negative values in what should have been NoData space (e.g., ocean). The pipeline successfully “healed” these regions by masking them to -32768, ensuring they do not bias modeling results or visual displays.
- **Metadata Mapping:** 100% of the 111 variables are successfully cross-referenced against the original Alaska Soil Data Bank (AKSDB) and Colby Brungard (2024) references, ensuring long-term traceability.

Scaling Quality Metrics

- **% Clamped:** Percentage of valid pixels saturated at Int16 limits (+/- 32,000).
- **Step/IQR:** Quantization step (1/scale) relative to Interquartile Range. Values < 0.1% indicate excellent precision.

Detailed Covariate Summary

Continuous Covariates Summary

Focal Statistics

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Deviance from Mean Elevation (Window: 4)	devmeanelev_4	_1k	10m	Direct	104.2	30.3	1,000	I16	0.00%	0.4214%
Deviance from Mean Elevation (Window: 16)	devmeanelev_16	_1k	10m	Direct	104.2	27.6	1,000	I16	0.00%	0.2346%
Deviance from Mean Elevation (Window: 32)	devmeanelev_32	_1k	10m	Direct	104.2	25.5	1,000	I16	0.00%	0.1778%
Difference from Mean Elevation (Window: 4)	diffmeanelev_4	_100	10m	Direct	104.7	23.6	100	I16	0.00%	2.4848%
Difference from Mean Elevation (Window: 16)	diffmeanelev_16	_100	10m	Direct	105.0	25.8	100	I16	0.00%	0.4842%
Difference from Mean Elevation (Window: 32)	diffmeanelev_32	_100	10m	Direct	104.9	26.5	100	I16	0.00%	0.1977%
Minimum Elevation (Window: 4)	minelev_4	_1	10m	Direct	81.1	5.9	1	I16	0.00%	0.1655%
Minimum Elevation (Window: 16)	minelev_16	_1	10m	Direct	66.6	4.4	1	I16	0.00%	0.1733%
Minimum Elevation (Window: 32)	minelev_32	_1	10m	Direct	56.3	3.4	1	I16	0.00%	0.1810%
Percentile Elevation (Window: 4)	perctelev_4	_100	10m	Semantic	84.7	30.4	100	I16	0.00%	0.0749%
Percentile Elevation (Window: 16)	perctelev_16	_100	10m	Semantic	91.0	31.2	100	I16	0.00%	0.0483%
Percentile Elevation (Window: 32)	perctelev_32	_100	10m	Semantic	92.0	30.7	100	I16	0.00%	0.0371%
Relative Elevation (Window: 4)	relelev_4	_10	10m	Semantic	95.7	15.5	10	I16	0.00%	0.9332%
Relative Elevation (Window: 16)	relelev_16	_10	10m	Semantic	95.1	16.4	10	I16	0.00%	0.2678%
Relative Elevation (Window: 32)	relelev_32	_10	10m	Semantic	94.7	16.5	10	I16	0.00%	0.1570%
Relative Mean Elevation (Window: 4)	relmeanelev_4	_100	10m	Semantic	104.3	23.6	100	I16	0.00%	2.4847%
Relative Mean Elevation (Window: 16)	relmeanelev_16	_100	10m	Semantic	104.6	25.8	100	I16	0.00%	0.4842%
Relative Mean Elevation (Window: 32)	relmeanelev_32	_100	10m	Semantic	104.6	26.5	100	I16	0.00%	0.1977%
stddelev (Window: 4)	stddelev_4	_100	10m	Semantic	95.6	19.4	100	I16	0.00%	0.1777%
stddelev (Window: 16)	stddelev_16	_100	10m	Semantic	95.4	16.7	100	I16	0.00%	0.0517%
stddelev (Window: 32)	stddelev_32	_100	10m	Semantic	95.1	14.7	100	I16	0.00%	0.0287%

Hydrology

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Catchment Area (Area: 10)	ca_10	_1	10m	Direct	100.0	74.3	1	F32	0.00%	Perfect

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Catchment Area (Area: 10000)	ca_10000	_1	10m	Direct	100.0	74.3	1	F32	0.00%	Perfect
Modified Catchment Area (Area: 10)	mca_10	_1	10m	Direct	99.9	72.6	1	F32	0.00%	Perfect
Modified Catchment Area (Area: 10000)	mca_10000	_1	10m	Direct	100.1	73.5	1	F32	0.00%	Perfect
Saga Wetness Index (Area: 10)	swi_10	_1k	10m	Direct	90.8	33.6	1,000	I16	0.00%	0.0143%
Saga Wetness Index (Area: 10000)	swi_10000	_1k	10m	Direct	91.0	34.0	1,000	I16	0.00%	0.0194%
Stream Power Index	spi	_10	10m	Direct	100.9	33.8	10	I16	6.61%	0.0127%
Topographic Wetness Index	twi	_1k	10m	Direct	89.6	34.3	1,000	I16	0.00%	0.0378%
Valley Depth	vlyd	_10	10m	Semantic	95.6	19.4	10	I16	0.00%	0.1111%

Lighting/Visibility

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Diurnal Anisotropic Heating	dah	_10k	10m	Direct	101.9	30.7	10,000	I16	0.00%	0.1698%
Hillshade (Standard)	hs_st	_100	10m	Direct	87.4	28.2	100	I16	0.00%	0.2669%
Topographic Openness (Negative) (Window: 2)	no_2	_10k	10m	Direct	76.3	29.2	10,000	I16	0.00%	1.0744%
Topographic Openness (Difference) (Window: 2)	diffopen_2	_100k	10m	Direct	90.2	37.5	100,000	I16	0.40%	0.0537%
Topographic Openness (Positive) (Window: 2)	po_2	_10k	10m	Direct	76.3	29.2	10,000	I16	0.00%	1.0744%
Topographic Openness (Negative) (Window: 32)	no_32	_10k	10m	Direct	78.2	28.8	10,000	I16	0.00%	0.3233%
Topographic Openness (Difference) (Window: 32)	diffopen_32	_100k	10m	Direct	89.2	37.0	100,000	I16	0.56%	0.0580%
Topographic Openness (Positive) (Window: 32)	po_32	_10k	10m	Direct	78.4	28.9	10,000	I16	0.00%	0.3201%
Topographic Openness (Difference) (Window: 256)	diffopen_256	_100k	10m	Direct	91.6	37.0	100,000	I16	1.17%	0.0343%
Topographic Openness (Negative) (Window: 256)	no_256	_10k	10m	Direct	80.4	28.0	10,000	I16	0.00%	0.1778%
Topographic Openness (Positive) (Window: 256)	po_256	_10k	10m	Direct	81.6	28.3	10,000	I16	0.00%	0.1708%
Potential Incoming Solar Radiation (Direct) (Date: 2023-01-22)	pisrdir_2023-01-22	_1k	10m	Direct	68.5	9.0	1,000	I16	0.00%	0.8315%
Potential Incoming Solar Radiation (Direct) (Date: 2023-02-22)	pisrdir_2023-02-22	_1k	10m	Direct	92.4	19.4	1,000	I16	0.00%	0.1986%
Potential Incoming Solar Radiation (Direct) (Date: 2023-03-22)	pisrdir_2023-03-22	_1k	10m	Direct	96.0	24.9	1,000	I16	0.00%	0.1097%

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Potential Incoming Solar Radiation (Direct) (Date: 2023-04-22)	pisrdir_2023-04-22	_1k	10m	Direct	93.9	26.3	1,000	I16	0.00%	0.1044%
Potential Incoming Solar Radiation (Direct) (Date: 2023-05-22)	pisrdir_2023-05-22	_1k	10m	Direct	91.0	25.9	1,000	I16	0.00%	0.1373%
Potential Incoming Solar Radiation (Direct) (Date: 2023-06-22)	pisrdir_2023-06-22	_1k	10m	Direct	89.7	25.2	1,000	I16	0.00%	0.1891%
Potential Incoming Solar Radiation (Direct) (Date: 2023-12-22)	pisrdir_2023-12-22	_1k	10m	Direct	45.4	4.8	1,000	I16	0.00%	5.3128%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-01-22)	pisrdif_2023-01-22	_10k	10m	Direct	79.3	10.5	10,000	I16	0.00%	0.0786%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-02-22)	pisrdif_2023-02-22	_10k	10m	Direct	80.8	14.8	10,000	I16	0.00%	0.0478%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-03-22)	pisrdif_2023-03-22	_10k	10m	Direct	80.5	17.5	10,000	I16	0.00%	0.0871%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-04-22)	pisrdif_2023-04-22	_10k	10m	Direct	78.6	18.3	10,000	I16	0.00%	0.1105%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-05-22)	pisrdif_2023-05-22	_10k	10m	Direct	79.8	19.5	10,000	I16	0.00%	0.0609%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-06-22)	pisrdif_2023-06-22	_10k	10m	Direct	80.3	19.9	10,000	I16	0.00%	0.0642%
Potential Incoming Solar Radiation (Diffuse) (Date: 2023-12-22)	pisrdif_2023-12-22	_10k	10m	Direct	60.9	7.5	10,000	I16	0.00%	0.0644%

Morphometry

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Aspect (Window: 4)	aspct_4	_10	10m	Direct	94.5	26.0	10	I16	0.00%	0.0630%
Aspect (Window: 16)	aspct_16	_10	10m	Direct	94.9	18.3	10	I16	0.00%	0.0626%
Aspect (Window: 32)	aspct_32	_10	10m	Direct	95.0	14.2	10	I16	0.00%	0.0622%
Convergence Index (Window: 4)	ci_4	_100	10m	Direct	104.1	34.4	100	I16	0.00%	0.0750%
Convergence Index (Window: 16)	ci_16	_100	10m	Direct	103.9	30.9	100	I16	0.00%	0.0467%
Convergence Index (Window: 32)	ci_32	_100	10m	Direct	103.7	27.5	100	I16	0.00%	0.0388%
Cross-sectional Curvature (Window: 4)	crosc_4	_1M	10m	Direct	104.7	32.5	1,000,000	I16	0.04%	0.2016%

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Cross-sectional Curvature (Window: 16)	crosc_16	_1M	10m	Direct	105.2	21.2	1,000,000	I16	0.00%	0.5245%
Cross-sectional Curvature (Window: 32)	crosc_32	_1M	10m	Direct	105.3	13.8	1,000,000	I16	0.00%	0.9536%
Longitudinal Curvature (Window: 4)	longc_4	_1M	10m	Direct	104.8	32.8	1,000,000	I16	0.04%	0.1774%
Longitudinal Curvature (Window: 16)	longc_16	_1M	10m	Direct	104.8	21.7	1,000,000	I16	0.00%	0.4624%
Longitudinal Curvature (Window: 32)	longc_32	_1M	10m	Direct	104.6	14.2	1,000,000	I16	0.00%	0.6710%
Mass Balance Index (Threshold: 0.001)	mbi_0.001	_10k	10m	Direct	99.0	39.4	10,000	I16	0.00%	0.0088%
Mass Balance Index (Threshold: 0.01)	mbi_0.01	_10k	10m	Direct	102.6	37.3	10,000	I16	0.00%	0.0415%
Mass Balance Index (Threshold: 0.1)	mbi_0.1	_100k	10m	Direct	104.0	38.3	100,000	I16	0.70%	0.0365%
Maximum Curvature (Window: 4)	maxc_4	_1M	10m	Semantic	102.6	32.2	1,000,000	I16	0.13%	0.1075%
Maximum Curvature (Window: 16)	maxc_16	_1M	10m	Semantic	103.5	20.1	1,000,000	I16	0.00%	0.2769%
Maximum Curvature (Window: 32)	maxc_32	_1M	10m	Semantic	103.8	12.5	1,000,000	I16	0.00%	0.4708%
Mid-slope Position	msp	_10k	10m	Direct	92.0	31.2	10,000	I16	0.00%	0.0172%
Minimum Curvature (Window: 4)	minc_4	_1M	10m	Semantic	103.0	32.5	1,000,000	I16	0.02%	0.1051%
Minimum Curvature (Window: 16)	minc_16	_1M	10m	Semantic	103.0	20.5	1,000,000	I16	0.00%	0.2467%
Minimum Curvature (Window: 32)	minc_32	_1M	10m	Semantic	102.7	12.7	1,000,000	I16	0.00%	0.3625%
Multiscale Topographic Position Index (Window: 4)	tpi_4	_1k	10m	Direct	103.8	33.9	1,000	I16	0.00%	0.1623%
Multiscale Topographic Position Index (Window: 32)	tpi_32	_1k	10m	Direct	104.4	32.8	1,000	I16	0.00%	0.1260%
Normalized Height	nh	_10k	10m	Semantic	92.7	28.8	10,000	I16	0.00%	0.0192%
Plan Curvature (Window: 4)	planc_4	_100k	10m	Direct	104.9	34.1	100,000	I16	0.09%	0.1222%
Plan Curvature (Window: 16)	planc_16	_100k	10m	Direct	105.5	24.7	100,000	I16	0.02%	0.2637%
Plan Curvature (Window: 32)	planc_32	_100k	10m	Direct	105.8	18.0	100,000	I16	0.02%	0.3950%
Profile Curvature (Window: 4)	profc_4	_1M	10m	Direct	104.8	32.5	1,000,000	I16	0.00%	0.1887%
Profile Curvature (Window: 16)	profc_16	_1M	10m	Direct	104.8	21.2	1,000,000	I16	0.00%	0.4835%
Profile Curvature (Window: 32)	profc_32	_1M	10m	Direct	104.7	13.9	1,000,000	I16	0.00%	0.6917%
Slope (Window: 4)	sl_4	_100	10m	Direct	96.0	23.6	100	I16	0.00%	0.0839%
Slope (Window: 16)	sl_16	_100	10m	Direct	96.4	15.0	100	I16	0.00%	0.0966%

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Slope (Window: 32)	sl_32	_100	10m	Direct	96.8	10.7	100	I16	0.00%	0.1107%
Slope Height	slh	_10	10m	Semantic	95.8	17.0	10	I16	0.00%	0.2930%
Standardized Height	stdh	_10	10m	Semantic	92.5	23.0	10	I16	0.00%	0.0296%
Terrain Ruggedness Index (Window: 4)	tri_4	_100	10m	Direct	95.4	20.9	100	I16	0.00%	0.1728%
Terrain Ruggedness Index (Window: 16)	tri_16	_100	10m	Direct	95.4	21.9	100	I16	0.00%	0.0479%
Terrain Ruggedness Index (Window: 32)	tri_32	_100	10m	Direct	95.3	22.4	100	I16	0.00%	0.0255%
Terrain Surface Convexity (Window: 4)	tsc_4	_100	10m	Direct	93.6	33.6	100	I16	0.00%	0.0345%
Terrain Surface Convexity (Window: 16)	tsc_16	_100	10m	Direct	91.9	19.8	100	I16	0.00%	0.0837%
Terrain Surface Convexity (Window: 32)	tsc_32	_100	10m	Direct	91.3	11.6	100	I16	0.00%	0.1218%
Vector Ruggedness Index (Window: 4)	vrn_4	_100k	10m	Semantic	99.8	20.8	100,000	I16	0.00%	0.4308%
Vector Ruggedness Index (Window: 16)	vrn_16	_100k	10m	Semantic	98.5	18.0	100,000	I16	0.07%	0.0970%
Vector Ruggedness Index (Window: 32)	vrn_32	_100k	10m	Semantic	97.5	15.4	100,000	I16	0.04%	0.0522%

Unknown

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
dfa	dfa	_10	10m	Unverified	201.2	34.1	10	I16	3.66%	0.0431%
dis	dis	_100	10m	Unverified	7.2	2.9	100	I16	0.00%	1.6111%
fel	fel	_1	10m	Unverified	190.8	6.5	1	I16	0.00%	0.1617%

Categorical Covariates Summary

Geomorphons

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Geomorphon (Multiscale) (Radius: 30)	gmrph_ms_30	_1	30m	Direct	0.5	0.6	1	Byte	0.00%	Perfect
Geomorphon (Radius) (Radius: 30)	gmrph_r_30	_1	30m	Semantic	0.5	0.5	1	Byte	0.00%	Perfect
Geomorphon (Radius) (Radius: 300)	gmrph_r_300	_1	30m	Semantic	0.5	0.6	1	Byte	0.00%	Perfect
Geomorphon (Multiscale) (Radius: 300)	gmrph_ms_300	_1	30m	Direct	0.5	0.6	1	Byte	0.00%	Perfect
Geomorphon (Radius) (Radius: 3000)	gmrph_r_3000	_1	30m	Semantic	0.5	0.6	1	Byte	0.00%	Perfect

Morphometry

Variable Name	Abbr	Suffix	Res	Match	Raw GB	Scaled GB	Scale	Type	% Clamp	Step/IQR
Morphometric Features (Window: 4)	morpfeat_4	_1	30m	Direct	0.6	0.7	1	Byte	0.00%	Perfect
Morphometric Features (Window: 16)	morpfeat_16	_1	30m	Direct	0.4	0.4	1	Byte	0.00%	Perfect
Morphometric Features (Window: 32)	morpfeat_32	_1	30m	Direct	0.3	0.3	1	Byte	0.00%	Perfect

File Size Comparison

Metric	Value
Total Raw Size	9.58 TB
Total Scaled Size	2.54 TB
Storage Reduction	73.5%

Final Verification

- **NoData Integrity:** 100% verified via 10,000-point GEE comparison.
- **Asset Sync:** All 111 bands registered in GEE (aksdb_topo_v20250422_scaled).